

# Data Mining for Business Analytics

## **MSBA 511**

Text Mining

# Introduction to Text Mining

Up until now we have only worked with **structured quantitative data** (e.g., numbers, binary/categorical variables). Now, we will shift our focus to **unstructured text data** such as:

- Tweets (Social media posts)
- Customer reviews
- Free-form response fields
- Medical records

**Goal:** Transform raw textual information and label document as belonging to a class or group similar text documents.



The broader term natural language processing (NLP) is more commonly used today with an emphasis on deriving understanding of the text and interpreting it like a human.

# Why Mine Text Data?

**Challenge:** Valuable insights often hidden in unstructured claim notes.



## CLAIM FORM AND INSTRUCTIONS

If you have any questions regarding benefits available, or how to file your claim, or if you would like to appeal any determination, please contact our Customer Care Center at 1-800-348-4489, 8:00 A.M. to 8:00 P.M. Eastern Standard Time

The furnishing of this form, or its acceptance by the Company as proof, must not be construed as an admission of any liability on the part of the Company, nor a waiver of any of the conditions of the insurance contract.

### INSTRUCTIONS FOR FILING YOUR GROUP ACCIDENT CLAIM

## **Approach:**

- Extract and transform text from claim forms into features.
- Train a predictive model using labeled past claims (fraudulent vs. non-fraudulent).
- Apply the model to flag suspicious new claims for review.

**Impact:** Enhances fraud detection by incorporating qualitative insights from written text.

# Why Mine Text Data?

**Challenge:** Clinics receive a high volume of online appointment requests written in free text.

✓ Department  
● Reason  
● Visit type  
● Location  
● Date and time  
● Book

What's the reason for your appointment?

Describe a symptom, condition, or reason for a visit

ENTER REASON (Required)

Example: I've had left knee pain for 3 months.  
300 of 300 characters left

Cancel Back Next

## Approach:

- Use text mining to extract key symptoms and urgency indicators from patient-submitted forms.
- Classify and route requests to the appropriate level of care:
  - **Administrative Assistant** (e.g., scheduling or insurance questions)
  - **Nurse** (e.g., minor issues or follow-ups)
  - **Doctor** (e.g., urgent or complex medical needs)

**Impact:** Streamlines triage, reduces delays, and ensures patients receive the right level of care promptly.

# Why Mine Text Data?

**Challenge:** Businesses receive large volumes of unstructured customer reviews.

## Approach:

- Use text mining to identify key themes, sentiments, and product/service feedback.
- Highlight what customers liked or disliked, and extract frequently mentioned features.
- Summarize or highlight helpful reviews to assist future buyers in decision-making

**Impact:** Enhances customer experience, supports product improvement, and aids potential customers with relevant **insights**.

### Customer reviews

★★★★☆ 4.6 out of 5

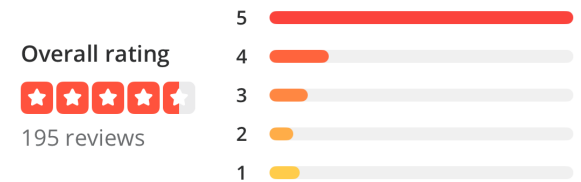
5,006 global ratings

### Customers say

Customers find the book excellent for developing data visualization strategies, with practical examples and easy-to-follow explanations. Moreover, the visual quality receives positive feedback for its illustrations on real examples, and customers appreciate how it delivers compelling data-driven stories while making the topic fun to read. Additionally, the book effectively eliminates clutter in charts. However, opinions about the book's content are mixed, with some finding it opinionated, and customers disagree on its value for money.

AI-generated from the text of customer reviews

### Recommended reviews



Q Search

Yelp sort

Filter by rating

### Insights

😊 Ambience (45)

😊 Drinks (8)

😊 Food (110)

😊 Service (71)

😊 Wait time (6)

# Key Steps in Text Mining

## 1. Text Collection

Gather unstructured data from sources like social media, emails, reports, reviews, etc.

## 2. Text Preprocessing

- Tokenization: Split text into words or terms
- Ignore case, stop word removal, lemmatization/stemming
- Noise removal: punctuation, numbers, special characters

## 3. Text Representation

Convert text into numerical format using:

- Bag-of-Words
- TF-IDF (Term Frequency-Inverse Document Frequency)

## 4. Modeling and Analysis

Apply data mining techniques such as clustering and classification.

# Bag-of-Words

The term Bag-of-Words (BoW) comes from the fact it treats text just as a collection (**bag**) of individual words:

- Ignores grammar, syntax, punctuation, and word order.
- Focuses on word frequency or presence, not structure

This technique is surprisingly effective for tasks like:

- Classification tasks especially for short-form text
- Sentiment identification for content such as customer reviews

Common in supervised learning with large text **corpora**.

- ✓ Does not require perfect accuracy to deliver valuable insights.



**Corpora often refers to a fixed standard set of text documents that many researches can use to develop and tune text mining algorithms.**

# Tokenization

The first step is to break the text up into individual terms. Tokenization is the process of identifying delimiters (e.g., spaces, punctuation) and separate the text into meaningful units called **tokens**.

For large text collections, tokenization can produce thousands to millions of unique terms. The English language alone has over 1 million words, not including misspellings, slang, or symbols.

## **Challenge:**

High dimensionality can hinder analysis and model performance.

## **Solution:**

Apply preprocessing techniques to reduce the number of terms and focus on the most informative tokens.



# Lemmatization vs. Stemming

Lemmatization and stemming are both techniques used to reduce words to their root form but differ in accuracy and linguistic awareness.

| Feature        | Stemming                                          | Lemmatization                                  |
|----------------|---------------------------------------------------|------------------------------------------------|
| Definition     | Remove suffixes to reduce words to base/root form | Convert words to their dictionary (lemma) form |
| Example        | "running" → "run"<br>"technical" → "technic"      | "running" → "run"<br>"better" → "good"         |
| Method         | Rule-based truncation                             | Dictionary and grammar-based lookup            |
| Speed          | Fast, simple                                      | Slower, more accurate                          |
| Output Quality | May create non-words                              | Produces valid words                           |
| Use Case       | Large-scale, quick preprocessing                  | NLP tasks needing higher linguistic accuracy   |

# Text Preprocessing Example

Let's practice and work through a simple text example:

*The technician was resolving technical issues quickly.*

**Step 1:** Tokenization based on the space delimiter:

["The", "technician", "was", "resolving", "technical", "issues", "quickly"]

**Step 2:** Ignore case and stop word removal

["technician", "resolving", "technical", "issues", "quickly"]

**Step 3:** Stemming

["technician", "resolv", "technic", "issu", "quick"]

# BoW Matrix Representation

- Columns are terms
- Rows are text/documents
- Cells indicate binary or frequency of terms in documents

Let's review the matrix representation for the following:

*S1: The technician was resolving technical issues quickly.*

*S2: The engineer resolved several technical problems efficiently.*

| Text | technician | resolv | technic | issu | quick | engin | sever | problem | effici |
|------|------------|--------|---------|------|-------|-------|-------|---------|--------|
| S1   | 1          | 1      | 1       | 1    | 1     | 0     | 0     | 0       | 0      |
| S2   | 0          | 1      | 1       | 0    | 0     | 1     | 1     | 1       | 1      |



**There are many different stemming algorithms (Porter, Snowball etc.), and they can produce different stems for the same word. Always choose a method that balances simplicity, accuracy, and consistency for your analysis.**

# TF-IDF Representation

Another popular option that factors in both the frequency of a term in the document as well as all documents with that term is **Term Frequency-Inverse Document Frequency (TF-IDF)**.

*$TF(t, d)$  = Number of times  $t$  appears in document  $d$*

The **Inverse Document Frequency (IDF)** for term  $t$  is calculated across all documents and defined as:

$$IDF(t) = 1 + \log\left(\frac{\text{Total number of documents}}{\text{Number of documents with } t}\right)$$

Finally, the **TF-IDF( $t$ ,  $d$ )** for a specific term-document pair is:

$$TF - IDF(t, d) = TF(t, d) \times IDF(t)$$

# Let's Practice: TF-IDF

*S1: The technician was resolving technical issues quickly.*

*S2: The engineer resolved several technical problems efficiently.*

| Text | technician | resolv | technic | issu | quick | engin | sever | problem | effici |
|------|------------|--------|---------|------|-------|-------|-------|---------|--------|
| S1   | 1          | 1      | 1       | 1    | 1     | 0     | 0     | 0       | 0      |
| S2   | 0          | 1      | 1       | 0    | 0     | 1     | 1     | 1       | 1      |

$$TF(technician, S1) = 1 \quad TF(technician, S2) = 0$$

$$IDF(technician) = 1 + \log\left(\frac{2}{1}\right) = 1.69$$

$$TF - IDF(technician, S1) = 1 \times 1.69 = 1.69$$

$$TF - IDF(technician, S2) = 0 \times 1.69 = 0$$

# Let's Practice: TF-IDF

*S1: The technician was resolving technical issues quickly.*

*S2: The engineer resolved several technical problems efficiently.*

| Text | technician | resolv | technic | issu | quick | engin | sever | problem | effici |
|------|------------|--------|---------|------|-------|-------|-------|---------|--------|
| S1   | 1          | 1      | 1       | 1    | 1     | 0     | 0     | 0       | 0      |
| S2   | 0          | 1      | 1       | 0    | 0     | 1     | 1     | 1       | 1      |

$$TF(resolv, S1) = 1 \qquad TF(resolv, S2) = 1$$

$$IDF(resolv) = 1 + \log\left(\frac{2}{2}\right) = 1.00$$

$$TF - IDF(resolv, S1) = 1 \times 1.00 = 1.00$$

$$TF - IDF(resolv, S2) = 1 \times 1.00 = 1.00$$

# Bag-of-Words vs. TF-IDF

Use Bag-of-Words when frequent terms matter (e.g., spam, sentiment clues) and TF-IDF when differentiation of text is important.

| Criteria         | Bag-of-Words                                | TF-IDF                                                       |
|------------------|---------------------------------------------|--------------------------------------------------------------|
| Best For         | Detecting patterns on frequent words        | Highlighting unique or distinguishing words                  |
| Best Use Cases   | Spam Detection, Sentiment Classification    | Document similarity, Topic Modeling, Content Recommendations |
| Frequent Words   | Tread as strong signal                      | Down weighted                                                |
| Rare Words       | May have little impact                      | Get higher weights and can be useful for differentiation     |
| Short Texts      | Often more effective due to limited context | May under weight rare but important terms                    |
| Long Texts       | May be overwhelmed by common words          | Handles length and vocabulary variation better               |
| Interpretability | Simple and intuitive                        | Requires understanding of log-weighting                      |